TinyDETR-Pose: Towards End-To-End Real-Time Single-Stage 6DoF Object Pose Estimation with Lightweight Transformers

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Abstract. Real-time 6DoF object pose estimation on resource-constrained hardware remains challenging, as accurate correspondence-based and refinement pipelines typically rely on non-differentiable PnP/RANSAC stages or costly iterative refinement, while recent foundation-model-based approaches incur inference costs that are prohibitive for edge deployment. We present **TinyDETR-Pose**, a lightweight, end-to-end, single-stage framework that jointly detects objects and regresses their full 6D pose in a single forward pass. Built on the efficient LW-DETR architecture, TinyDETR-Pose formulates detection and pose estimation as a set-prediction problem and attaches dedicated MLP heads for rotation, monocular depth, and projected object center regression to each decoder query, eliminating the need for PnP, NMS (non-maximum suppression), or iterative pose refinement. Object symmetries are handled through a single symmetry-agnostic ADD-S loss applied uniformly to all objects, avoiding symmetry labels, object-specific loss schedules, or separate geodesic/ADD supervision. In addition, predictions are assigned to ground truth using a symmetry-safe Hungarian matcher based on class and 2D spatial cues, yielding stable assignment under symmetry and depth ambiguity. On YCB-V, TinyDETR-Pose achieves a comparable AUC of ADD-S of **85.9** and obtains the best ADD(-S) accuracy at the $0.1d$ threshold with **93.6**, while requiring $\approx$ **3.5**$\times$ fewer parameters than other DETR-based single-stage pose-estimation approaches. Due to its compact design, TinyDETR-Pose runs in real time and achieves an inference latency of only ~**4.5** ms per frame on an NVIDIA Jetson Nano using TensorRT, demonstrating that accurate end-to-end transformer-based 6D pose estimation can be made practical for edge deployment.

Keywords: 6DoF pose estimation · single-stage · detection transformer · real-time · edge deployment

1 Introduction

Accurately estimating an object's 6DoF (six degrees of freedom) pose, meaning its full 3D rotation and translation relative to the camera, is essential in various fields such as robotics and augmented reality [25, 39]. Despite remarkable progress, there is still a basic trade-off between the accuracy and efficiency of these methods, which makes it challenging to deploy them in real-time systems [44]. The baseline methods rely on multi-stage pipelines that establish dense 2D–3D correspondences and recover the pose via RANSAC-based PnP solvers [30, 31, 43, 47], making them non-differentiable and expensive. Direct regression approaches [6, 20, 40, 42] bypass PnP but still depend on a separate 2D detector, preserving a two-stage structure. Foundation models [17, 27, 29, 41] achieve impressive zero-shot generalization, but they have inference times not tailored for real-time systems, i.e. FoundationPose [41] requires $\sim$1.3 s per object on high-end desktop GPUs, precluding edge deployment. This performance limitation motivates single-stage architectures that jointly detect all objects and estimate their 6D poses in a single forward pass [1, 13, 24, 32]. However, existing single-stage methods struggle with monocular depth ambiguity, lack principled symmetry handling, and provide limited rotation supervision.

We present **TinyDETR-Pose**, an end-to-end, single-stage 6D pose estimation framework built on the lightweight detection transformer LW-DETR [5] compatible with edge devices. We treat joint detection and pose estimation as a set-prediction problem [1, 4], regressing each instance's full 6D pose directly from the decoder output via dedicated MLP heads for rotation, monocular depth, and geometric keypoints—achieving inference in 4.5 ms on Jetson Nano.

Our main contributions are:

- A single-stage, query-based architecture that jointly detects objects and regresses their full 6D pose without PnP, NMS, or iterative refinement.
- A unified, symmetry-agnostic ADD-S training objective that supervises the complete pose without requiring symmetry labels, geodesic losses, or per-object loss curricula.
- A projected-center-based bipartite matcher that excludes rotation and depth from the assignment cost, ensuring stable matching under symmetry and depth ambiguity.

We evaluate on YCB-V [42], comparing against single-stage methods [1, 13, 24, 32] and direct-regression baselines [40, 42].

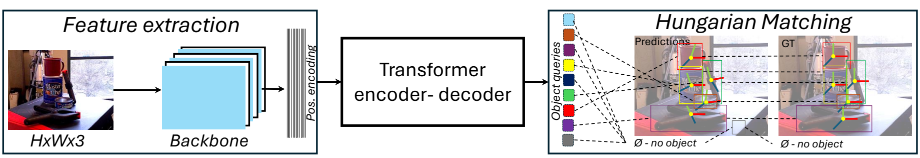

Fig. 1: Overview of TinyDETR-Pose. Given a single RGB image, our model jointly detects all objects and regresses their full 6D pose in one forward pass via set prediction; see Fig. 2 for architectural details.

2 Related Work

Deep learning approaches to 6D object pose estimation broadly fall into three categories: indirect correspondence-based methods, direct regression frameworks, and foundation models. Orthogonally, DETR-style set-prediction architectures have been extended beyond 2D detection to end-to-end 6D pose estimation, motivating our query-based formulation.

2.1 Indirect and Direct Regression Methods

Indirect methods establish 2D–3D correspondences and recover the pose via PnP [47]. BB8 [33] regresses projections of bounding-box corners; PVNet [31] improves occlusion robustness through pixel-wise voting with RANSAC [7]; DPOD [43] and Pix2Pose [30] extend this to dense UV and per-pixel 3D coordinate maps, respectively.While robust, these pipelines remain multi-stage and are typically not end-to-end differentiable due to the PnP/RANSAC step. As a result, intermediate prediction errors can propagate to the final pose, the correspondence estimator cannot be trained directly from the final pose objective, and the overall pipeline depends on additional hand-tuned components such as RANSAC thresholds.
Direct methods regress object rotation and translation directly from the image. PoseCNN [42] regressed quaternions but suffered from representation discontinuities later addressed by the continuous 6D parameterization of Zhou et al. [48]. CDPN [20] disentangles rotation and translation into separate branches, and GDR-Net [40] bridges direct and indirect paradigms by guiding regression with dense correspondence maps. SO-Pose [6] adds self-occlusion cues for complementary supervision, while CosyPose [16] extends the paradigm to multi-object scenes via iterative render-and-compare refinement, achieving top results on the BOP benchmark [11]. Despite their accuracy, most direct methods still rely on a separate 2D detector [20, 40], making them two-stage at inference.

2.2 Foundation Models for Unseen Objects

Recent zero-shot 6D pose estimation methods aim to improve generalization by leveraging synthetic rendering, template matching, and foundation-model features. MegaPose [17] introduced a render-and-compare strategy for zero-shot pose estimation but requires over 15 s per object. GigaPose [27] achieves a 35× speedup through discriminative templates, yet still needs several hundred milliseconds. FoundationPose [41] combines large-scale synthetic training with neural implicit representations for state-of-the-art BOP results [11], and FoundPose [29] shows that frozen DINOv2 [28] features suffice for synthetic-to-real matching without task-specific training. Despite their generalization capabilities, these models are prohibitively heavy for edge devices e.g. FoundationPose requires ∼1.3 s per object even on high-end desktop GPUs.

2.3 Single-Stage and Transformer-based Approaches

Single-stage architectures unify detection and pose regression in one forward pass. YOLO-6D-Pose [24] extends YOLOX [8] to infer all poses simultaneously. EfficientPose [2] adapts EfficientDet [38] for scalable end-to-end estimation. Transformer-based variants further improve accuracy through global context: T6D-Direct [1] treats multi-object pose estimation as set prediction, PoET [13] leverages multi-scale features from pre-trained detectors, and YOLOPoseV2 [32] combines keypoint regression with a learnable rotation module to eliminate non-differentiable PnP solvers at real-time speed. However, such transformer-based designs typically require substantially larger models, with approximately 3.5× more parameters than our approach. Our work builds upon the single-stage paradigm from an efficiency-oriented perspective, achieving comparable accuracy with a parameter budget close to NMS-based CNN methods [24], making it suitable for edge-level compute constraints.

3 Methodology

In this section, we present **TinyDETR-Pose**, an end-to-end 6D object pose estimation pipeline suitable for edge devices. While building our approach upon LW-DETR [5], we introduce three methodological extensions for 6D object pose retrieval: (i) dedicated pose prediction heads for translation and rotation estimation, (ii) a pose-aware loss formulation for supervising 6D pose predictions, and (iii) a 2D center-based matching strategy that incorporates the projected object center (u, v) into the Hungarian assignment to improve pose-query association while remaining robust to symmetry and depth ambiguity. An architectural overview is provided in Fig. 2.

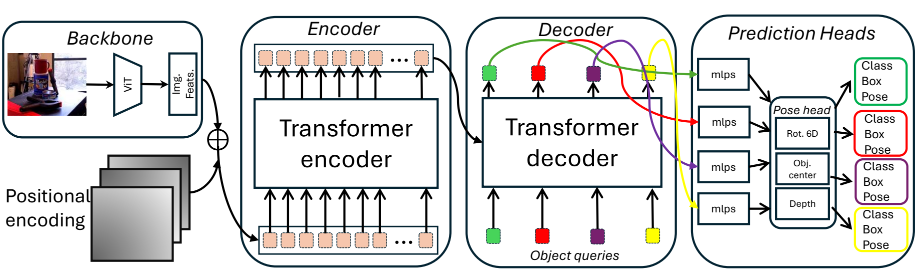

Fig. 2: Architecture of TinyDETR-Pose. A CAEv2-pretrained ViT backbone [45] feeds multi-scale features into a 3-layer transformer decoder attending over N learnable object queries. Hungarian matching [15] assigns predictions to ground truth during training, using our projected object-center cost (Sec. 3.1). Each matched query is decoded by shared, parallel MLP heads for class, box, rotation, and translation, see Sec. 3.2 for details on the translation decomposition and rotation parameterization.

3.1 6D Object Pose Estimation as Set Prediction

Inspired by the set-based prediction paradigm of DETR [4] and its extension to 6DoF pose estimation in T6D-Direct [1], we jointly regress 2D object detections and their corresponding 6D poses in a single forward pass. Building upon LW-DETR [5], our method inherits its lightweight design, making it particularly suited for edge deployment. In contrast to prior transformer-based pose estimation methods [1, 13] that regress the full translation vector directly, we decompose translation estimation into two distinct sub-problems [36]: instead of directly regressing the full metric translation vector, we estimate the projected object center in normalized UV space and the depth t_z separately. The lateral translation components (t_x, t_y) are then recovered using the camera intrinsic matrix $\mathbf{K}$. This factorization separates image-plane localization from monocular depth estimation, reducing the coupling between lateral position and scale/depth and leading to a more stable and camera-aware regression problem.

Set Prediction. Given an input image of size $(H \times W \times 3)$ our model predicts a fixed-size set of N tuples $\{\hat{q}_i\}_{i=1}^{N}$. Each tuple is represented as $\hat{q}_i = \big(\gamma_i,\, \mathbf{b}_i,\, \mathbf{uv}_{\text{cntr}_i},\, t_{z_i},\, \mathbf{r}_{6\text{D}_i}\big)$ comprising class label probabilities (γ_i), a bounding box $(\mathbf{b}_i)$, and a 6DoF object pose split into translation $(\mathbf{uv}_{\text{cntr}_i},\, t_{z_i})$ and a 6D rotation vector $(\mathbf{r}_{6\text{D}_i})$. All inputs but the class labels, are all normalized to $[0, 1]$. For rotation, we adopt the continuous 6D representation of Zhou et al. [48], which has been shown to yield superior training stability and downstream performance. Following the set prediction formulation of DETR [4], the decoder operates on N learnable object queries, where N is a hyperparameter set to exceed the maximum number of objects expected in any given image. Predictions not matched to a ground-truth target are supervised against a dedicated $\varnothing$ (*no-object*) class, enabling the model to explicitly learn to suppress false positives.

Enhanced Bipartite Matching. Following DETR [4], we use Hungarian matching to assign each predicted query $\hat{q}_i$ to a ground-truth instance y_j by minimizing a pairwise cost matrix $\mathcal{C}(\hat{q}_i, y_j)$ composed of class confidence $(\mathcal{C}_{\text{cls}})$, bounding-box ℓ_1 distance $(\mathcal{C}_{\text{bbox}})$, and IoU $(\mathcal{C}_{\text{IoU}})$. We additionally introduce a center cost $(\mathcal{C}_{\text{cntr}})$, defined as the ℓ_1 distance between predicted and ground-truth 3D object centers projected onto the image plane.

Final matching cost. We define our matching cost as:

$$\mathcal{C}(\hat{q}_i, y_j) = \lambda_{\text{cls}}\, \mathcal{C}_{\text{cls}} + \lambda_{\text{bbox}}\, \mathcal{C}_{\text{bbox}} + \lambda_{\text{IoU}}\, \mathcal{C}_{\text{IoU}} + \lambda_{\text{cntr}}\, \mathcal{C}_{\text{cntr}}, \tag{1}$$

Because rotation and depth are inherently ambiguous, we supervise them only through the post-matching loss, with symmetry-aware loss branches explicitly accounting for these ambiguities. This design is consistent with the matching philosophy of DETR [4] and its successors [5, 18, 21, 26, 37, 46], which rely exclusively on 2D geometric quantities in the assignment cost to ensure stable convergence, deferring task-specific supervision entirely to the training loss. We set the matching cost weights in $\mathcal{C}(\hat{q}_i, y_j)$ to:

$$(\lambda^{\mathcal{C}}_{\text{cls}},\, \lambda^{\mathcal{C}}_{\text{bbox}},\, \lambda^{\mathcal{C}}_{\text{IoU}},\, \lambda^{\mathcal{C}}_{\text{cntr}}) = (2.0,\, 5.0,\, 2.0,\, 5.0) \tag{2}$$

The $\lambda^{\mathcal{C}}_{\text{cls}}$, $\lambda^{\mathcal{C}}_{\text{bbox}}$, and $\lambda^{\mathcal{C}}_{\text{IoU}}$ weights follow the default matching configuration of LW-DETR [5]. The center term is specific to our pose-estimation formulation and was chosen empirically; setting $\lambda^{\mathcal{C}}_{\text{cntr}} = 5.0$ balances the center localization cost with the bounding-box regression cost and yielded stable matching in our experiments.

3.2 Model Architecture & Training Objective

In the following, we detail the network architecture, the design of our pose estimation head, and the training strategy including our proposed loss for 6DoF object pose regression.

Network architecture. Our pose estimation model builds upon the 2D detection framework LW-DETR [5]. We retain its default configuration, including the CAEv2 [45] autoencoder backbone, the plain ViT encoder stack, the projector structure, and a shallow 3-layer decoder, preserving its efficiency. Specifically, we adopt the same number of encoder and decoder layers, attention heads, hidden dimensions, and feed-forward network settings as LW-DETR. Our pose estimation head comprises three MLPs that share the same input dimension (d_h) and a single hidden layer, differing only in their output dimensions, as detailed below.

Pose estimation head design. All 6DoF pose parameters are predicted by three lightweight MLPs (Fig. 3), each with input dimension d_h=256 and a single hidden layer. The rotation head outputs a 6-dimensional vector using the continuous 6D parameterization [48], from which the rotation matrix is recovered by orthogonalization. For translation, following Simonelli et al. [36], we decompose the 3D translation vector into two complementary subproblems instead of directly regressing it in camera coordinates. The projected object center (u, v) is estimated in image space and supervised as a keypoint prediction task, while the metric depth t_z is treated as monocular depth estimation. This decomposition separates image-plane localization from depth regression and allows each subproblem to be supervised with an appropriate loss. For each matched instance i, let $\hat{\mathbf{c}}_i = (\hat{u}_i, \hat{v}_i)^\top$ denote the predicted projected object center, and let $\mathbf{c}_i = (u_i, v_i)^\top$ be the ground-truth center obtained by projecting the 3D object center into the image using the camera intrinsics $\mathbf{K}$. We supervise the projected center using an OKS-based loss [24]:

$$\text{OKS}_i = \exp\left(-\frac{\|\hat{\mathbf{c}}_i - \mathbf{c}_i\|_2^2}{2s_i^2\sigma_{\text{cntr}}^2}\right), \qquad \mathcal{L}_{\text{cntr}} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \text{OKS}_i\right), \tag{3}$$

where $s_i = \sqrt{w_i h_i}$ denotes the object scale from the ground-truth bounding box and σ_{cntr} controls the center-keypoint tolerance. For metric depth, we regress t_z in log-space:

$$e_i^z = \log \hat{t}_{z,i} - \log t_{z,i} = \log\left(\frac{\hat{t}_{z,i}}{t_{z,i}}\right), \tag{4}$$

$$\mathcal{L}_{t_z} = \frac{1}{N}\sum_{i=1}^{N} \text{Smooth}_{\ell_1}^{\beta}(e_i^z), \tag{5}$$

with

$$\text{Smooth}_{\ell_1}^{\beta}(e) = \begin{cases} \frac{1}{2\beta}e^2, & \text{if } |e| < \beta, \\ |e| - \frac{\beta}{2}, & \text{otherwise.} \end{cases} \tag{6}$$

At inference, the full translation (t_x, t_y, t_z) is recovered by back-projecting the predicted center (u, v) through $\mathbf{K}^{-1}$ at the predicted depth t_z.

Training strategy & loss. TinyDETR-Pose is a single-stage model that jointly optimizes detection and 6D pose. For detection, we use an IoU-aware classification loss $\mathcal{L}_{\text{cls}}$ (IA-BCE [3]), an ℓ_1 bounding-box loss $\mathcal{L}_{\text{bbox}}$, and a generalized IoU loss $\mathcal{L}_{\text{IoU}}$ [34]. For pose estimation, we use the center loss $\mathcal{L}_{\text{cntr}}$, the log-depth loss $\mathcal{L}_{t_z}$, and a geometry-grounded ADD-S loss. Rotation receives no direct component-wise supervision; instead, it is optimized implicitly through the ADD-S objective. Because ADD-S measures the distance to the nearest transformed model point, it provides a single pose-level signal for both symmetric and asymmetric objects without requiring symmetry labels. Let $\mathcal{M}_i = \{\mathbf{x}_{ij}\}_{j=1}^{M_i}$ denote the set of 3D model points for object instance i; for each object mesh, we uniformly sample 512 points to construct $\mathcal{M}_i$. Given the predicted pose $(\hat{\mathbf{R}}_i, \hat{\mathbf{t}}_i)$ and ground-truth pose $(\mathbf{R}_i, \mathbf{t}_i)$, we define:

$$\hat{\mathbf{p}}_{ij} = \hat{\mathbf{R}}_i \mathbf{x}_{ij} + \hat{\mathbf{t}}_i, \qquad \mathbf{p}_{ij} = \mathbf{R}_i \mathbf{x}_{ij} + \mathbf{t}_i.$$

We use the symmetry-agnostic ADD-S distance for every instance:

$$d_i = \frac{1}{M_i} \sum_{k=1}^{M_i} \min_{j \in \{1,\ldots,M_i\}} \left\| \hat{\mathbf{p}}_{ij} - \mathbf{p}_{ik} \right\|_2 .$$

The final ADD-S loss over all matched instances is:

$$\mathcal{L}_{\text{ADD-S}} = \frac{1}{N} \sum_{i=1}^{N} d_i. \tag{7}$$

We initialize the backbone with weights pretrained on Objects365 [35] and use the following loss weight configuration: $\lambda_{\text{cls}}^{\mathcal{L}} = 1.0$, $\lambda_{\text{bbox}}^{\mathcal{L}} = 5.0$, $\lambda_{\text{IoU}}^{\mathcal{L}} = 2.0$, $\lambda_{\text{cntr}}^{\mathcal{L}} = 2.5$, $\lambda_{\text{ADD-S}}^{\mathcal{L}} = 30.0$, and $\lambda_{t_z}^{\mathcal{L}} = 15.0$. Combining the loss criteria leads us to the following equation:

$$\begin{aligned} \mathcal{L}_{\text{total}} = \lambda_{\text{cls}}^{\mathcal{L}} \cdot \mathcal{L}_{\text{cls}} + \lambda_{\text{bbox}}^{\mathcal{L}} \cdot \mathcal{L}_{\text{bbox}} + \lambda_{\text{IoU}}^{\mathcal{L}} \cdot \mathcal{L}_{\text{IoU}} \\ + \lambda_{\text{cntr}}^{\mathcal{L}} \cdot \mathcal{L}_{\text{cntr}} + \lambda_{t_z}^{\mathcal{L}} \cdot \mathcal{L}_{t_z} + \lambda_{\text{ADD-S}}^{\mathcal{L}} \cdot \mathcal{L}_{\text{ADD-S}} \end{aligned} \tag{8}$$

We train on with a batch size of 32, for 60 epochs. We use AdamW [22] with a learning rate of $1e^{-4}$. N is set to 100.

4 Experiments

We first describe the evaluation datasets and metrics, then compare our method against state-of-the-art baselines and analyze the impact of key design choices through ablation studies.

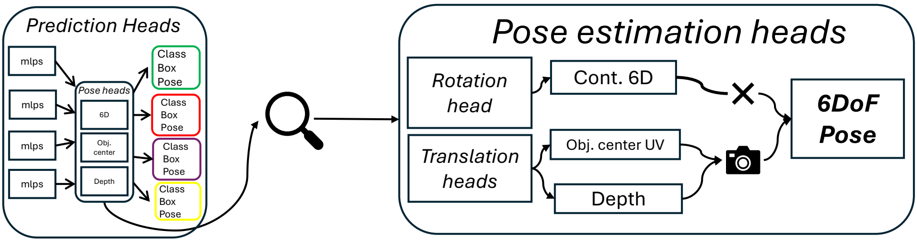

Fig. 3: Pose estimation head design. Rotation is predicted in the continuous 6D representation [48], with the full 3×3 matrix recovered via cross-product orthogonalization. For translation, following [36], the object center (t_x, t_y) is projected into normalized UV space using $\mathbf{K}$, while t_z is treated as monocular depth estimation. The predicted UV coordinates are directly used in the Hungarian matcher. The full 3D translation is recovered by back-projecting the UV center and t_z through $\mathbf{K}$, yielding the complete 6DoF pose.

4.1 Setup

We evaluate on YCB-V [42], a standard 6D pose benchmark of 21 household objects across 92 RGB-D sequences (640×480), annotated with ground-truth 6D poses, segmentation masks, depth, and bounding boxes. Following the standard protocol [42], we train on the training sequences plus 80K synthetic images [42] and evaluate on the 2,949 keyframes from the 12 test sequences. For validation we use the BOP subset of YCB-V [12] (75 frames per test scene, 900 images total). We assess performance with complementary metrics [1,10,13] that jointly capture translation, rotation, and overall pose quality: ADD-(S) at the $0.1d$ threshold, the AUC of ADD(-S), the AUC of ADD-S.

4.2 Comparison with state-of-the-art baselines

We compare TinyDETR-Pose against representative methods across three tiers: single-stage real-time methods, direct regression baselines, and refinement-based upper bounds. Tab. 1 reports results on the YCB-V benchmark [42]. Our method attains an AUC of ADD-S comparable to the state of the art while using only 14.5M parameters. In contrast, DETR-based approaches that reach state-of-the-art accuracy rely on substantially larger parameter budgets, primarily owing to their deeper decoder stacks and larger backbones. Despite this reduction in model size, the gap between our model and the state of the art on AUC of ADD-S remains small (only 5%). Moreover, by achieving 93.5% our model surpasses the only other method that reports ADD-S at the $0.1d$ threshold—YOLO-6D-Pose (88.5 %)—by a clear margin. As this metric measures the fraction of predicted poses whose average model-point distance falls within 10% of the object diameter, this result indicates that our model reliably yields poses that are accurate enough for downstream tasks such as robotic grasping and manipulation, exhibiting few gross pose errors. Notably, the CNN-based approach YOLO-6D-Pose (s) [24] achieves state-of-the-art AUC of ADD(-S) with as few as 11.6M

Method	RGB-only	Refinement	Real-time	Params (M)	Num. Decoders	ADD(-S) 0.1d (%) ↑	AUC of ADD(-S) ↑	AUC of ADD-S ↑
Refinement-based								
CosyPose (ECCV'20) [16]	✓	✓	✗	–	–	–	84.5	89.8
DeepIM (ECCV'18) [19]	✓	✓	✗	–	–	–	81.9	88.1
Direct regression								
GDR-Net (CVPR'21) [40]	✓	✗	✓	–	–	60.1	84.4	89.1
SO-Pose (ICCV'21) [6]	✓	✗	✓	–	–	56.8	83.9	90.9
PoseCNN (RSS'18) [42]	✓	✗	✓	–	–	–	75.9	61.3
Single-stage								
T6D-Direct$^\diamond$ (GCPR'21) [1]	✓	✗	✓	–	6	–	74.6	86.2
PoET$^\diamond$ (CoRL'22) [13]	✓	✗	✓	–	5	–	–	87.1
YOLO-6D-Pose (s)(3DV'24) [24]	✓	✗	✓	11.6	–	88.5	**84.4**	–
YOLOPoseV2$^\diamond$ (RAS'23) [32]	✓	✗	✓	48.6	6	–	82.6	90.1
YOLOPoseV2-A$^\diamond$ (RAS'23) [32]	✓	✗	✓	53.2	6	–	83.3	**91.2**
TinyDETR-Pose$^\diamond$ (Ours)	✓	✗	✓	14.5	3	**93.5**	65.6	85.9

Table 1: Comparison on YCB-V for 6D object pose estimation. Methods marked with $\diamond$ denote single-stage DETR-based approaches; for these methods, we also report the number of decoder layers and if reported the number of model parameters. Our model achieves the best ADD(-S) performance at the $0.1d$ threshold and obtains competitive AUC of ADD-S results, while requiring approximately 3.5× fewer parameters than other DETR-based methods.

parameters, yet it relies on NMS as a hand-crafted post-processing step and is therefore not truly end-to-end. Competing single-stage transformer-based methods, in turn, tend to struggle on this metric, typically requiring roughly 5× more parameters to reach comparable accuracy [23,32]. Our method departs from this trend: with only 14.5M parameters, it nearly matches the compactness of CNN-based models while retaining the advantages of a fully end-to-end, NMS-free transformer detector that directly predicts a set of poses. Its accuracy on the AUC of ADD(-S) nonetheless leaves some headroom, which we attribute to fine-grained pose precision and analyze in detail in our ablation studies.

Qualitative results. Fig. 4 (a,b,d,e) shows successful predictions of TinyDETR-Pose. The model aligns the predicted poses well with the ground truth even in cluttered scenes and handles both small and large objects reliably. Fig. 4 (c,f) depicts representative failure cases. We assume these errors are mainly caused by strong occlusion and partial visibility, especially for the scissors, where large parts of the object geometry are hidden and the pose becomes ambiguous.

Effect of the ADD-S Loss Weight. Tab. 2 shows that the $\lambda^{\mathcal{L}}_{\text{ADD-S}}$ weight has a clear impact on pose-estimation performance. Increasing $\lambda^{\mathcal{L}}_{\text{ADD-S}}$ from 10.0 to 20.0 leads to a gradual improvement in both AUC of ADD-S and AUC of ADD(-S), indicating that stronger geometric supervision is beneficial. The best overall performance is obtained with $\lambda^{\mathcal{L}}_{\text{ADD-S}} =$ 30.0, achieving the highest AUC of ADD-S, the highest AUC of ADD(-S). Fur-

$\lambda^{\mathcal{L}}_{\text{ADD-S}}$	AUC of ADD-S ↑	AUC of ADD(-S) ↑
10.0	79.5	57.1
20.0	81.0	58.5
30.0	**85.9**	**64.6**
50.0	83.1	59.4
75.0	78.63	57.5

Table 2: Ablation on the ADD-S loss coefficient $\lambda^{\mathcal{L}}_{\text{ADD-S}}$.

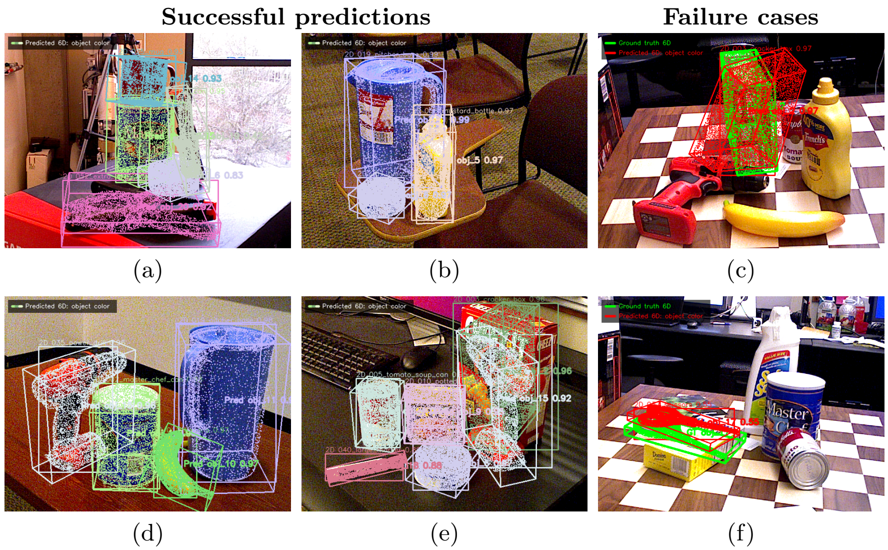

Fig. 4: Qualitative results of **TinyDETR-Pose** on YCB-V [42]. Each row shows two successful predictions followed by one representative failure case. Failures cases: Green denotes ground-truth poses, and red denotes predictions from our model. The failure cases are mainly caused by strong occlusion.

ther increasing the coefficient to 50.0 or 75.0 degrades the AUC metrics, suggesting that an overly large ADD-S loss weight disturbs the balance between the loss terms. Therefore, we use $\lambda^{\mathcal{L}}_{\text{ADD-S}} = 30.0$ as it provides the best trade-off across the evaluated metrics.

Matcher analysis. In the ablation of the projected object center matching cost $\mathcal{C}_{\text{cntr}}$ (see Tab. 3), we observe that adding the center term to the matching cost improves pose accuracy over matching on class and bounding-box cues alone.

Matcher	$\lambda^{\mathcal{C}}_{\text{cntr}}$	**AUC of ADD-S** ↑	**AUC of ADD(-S)** ↑
Default	0.0	84.5	63.1
+ $\mathcal{C}_{\text{cntr}}$ **(ours)**	**5.0**	**85.9**	**64.6**

Table 3: Ablation of the projected object center matching cost $\mathcal{C}_{\text{cntr}}$.

Runtime analysis. Tab. 4 reports the measured inference latency of the proposed model variants on a consumer-grade NVIDIA RTX 4090 GPU. TinyDETR-Pose is designed to address the efficiency requirements of edge deployment. It achieves a per-frame latency of approximately 9 ms in a PyTorch runtime environment. To further assess the edge-deployment capability of our method, we deploy our model on an NVIDIA Jetson Nano using TensorRT-optimized weights. The optimized model attains an inference latency of approximately 4.5 ms per frame while largely preserving its pose-estimation accuracy, demonstrating the suitability of our approach for real-time edge applications. TinyDETR-Pose

Method	Backbone	Num. Decoders	ms/frame ↓	AUC of ADD-S ↑
CosyPose† [16]	EfficientNet-B3	–	395	89.5
GDR-Net‡ [40]	ResNet-34	–	65	89.1
YOLO-6D-Pose (s)* [24]	CSPDarkNet-s	–	16.9	–
T6D-Direct [1]	ResNet-50	6	17	86.2
YOLOPoseV2 [32]	ResNet-50	6	17	90.1
YOLOPoseV2-A [32]	ResNet-50	6	22	91.2
PoET [13]	Scaled-YOLOv4	5	–	87.1
TinyDETR-Pose (ours)	ViT-Tiny	**3**	**9**	**85.9**
TinyDETR-Pose (TRT) (ours)	ViT-Tiny	**3**	**4.5**	**85.3**

Table 4: Runtime comparison on YCB-V. †: refinement-based, ‡: direct-regression, ∗: NMS post-processing; all others are single-stage. For DETR-based methods, decoder layers are reported. Latency is given in ms/frame. TinyDETR-Pose maintains comparable ADD-S AUC with substantially lower latency, further improved by TensorRT deployment on the NVIDIA Jetson Nano.

achieves a comparable AUC of ADD-S while running about **47**% faster then any other evaluated model; when additionally deployed through the TensorRT runtime, we observe a **73.5**% reduction in inference latency without a notable loss in performance.

Per object analysis. Tab. 5 provides a detailed per-object comparison on YCB-V benchmark dataset [42]. Overall, TinyDETR-Pose achieves competitive ADD-S performance on several mobjects despite its lightweight design, and obtains the best result for the *large marker*. In addition, our model performs on par with the strongest competing methods for objects such as *master chef can*, *tomato soup can*, *tuna fish can*, *potted meat can*,*foam brick* and *large marker*. However, the ADD(-S) results reveal larger performance gaps for some asymmetric objects, most notably the *cracker box* and *scissors*. We attribute this primarily to heavy occlusions in these instances, compounded by the limited refinement capacity of our shallow three-layer decoder. Increasing decoder depth or adding occlusion-aware reasoning could address this gap.

5 Conclusion

We presented **TinyDETR-Pose**, a lightweight end-to-end framework for single-stage 6D object pose estimation. By building on LW-DETR and formulating joint object detection and pose estimation as a set-prediction problem, TinyDETR-Pose directly regresses rotation, projected object center, and metric depth from decoder queries, without requiring PnP, NMS, or iterative pose refinement. The proposed symmetry-agnostic ADD-S supervision further allows a unified training objective for both symmetric and asymmetric objects, removing the need for explicit symmetry labels or object-specific pose losses. Experiments on YCB-V

Metric	AUC of ADD-S					AUC of ADD(-S)				
Object	**TinyDETR-Pose** (ours)	**PoET** [13]	**YOLO Pose** [32]	**YOLO Pose-A** [32]	**T6D** [1]	**TinyDETR-Pose** (ours)	**PoET** [13]	**YOLO Pose** [32]	**YOLO Pose-A** [32]	**T6D** [1]
master chef can	90.3	88.4	91.3	**91.7**	91.1	64.6	-	64.0	**71.3**	61.5
cracker box	78.5	80.5	86.8	**92.0**	86.6	6.5	-	77.9	**83.3**	76.3
sugar box	83.6	92.4	**92.6**	91.5	90.3	67.1	-	**87.3**	83.6	81.8
tomato soup can	89.7	**91.4**	90.5	87.8	88.9	64.7	-	**77.8**	72.9	72.0
mustard bottle	88.8	91.7	93.6	**96.6**	94.7	62.8	-	87.9	**93.4**	85.7
tuna fish can	93.3	90.4	94.3	**94.9**	92.2	50.3	-	**74.4**	70.5	59.0
pudding box	85.0	89.0	92.3	**92.6**	85.1	61.4	-	**87.9**	87.0	72.7
gelatin box	88.3	91.7	90.1	**92.2**	86.9	73.1	-	83.4	**85.7**	74.4
potted meat can	90.1	**91.2**	85.8	85.0	83.5	62.7	-	**76.7**	71.4	67.8
banana	87.7	89.5	95.0	**95.8**	93.8	46.1	-	88.2	**90.0**	87.4
pitcher base	90.4	91.7	93.6	**95.2**	92.3	68.2	-	88.5	**90.8**	84.5
bleach cleanser	79.5	**85.4**	85.3	83.1	83.0	53.2	-	**73.0**	70.8	65.0
bowl*	83.8	90.5	92.3	**93.4**	91.6	83.8	-	92.3	**93.4**	91.6
mug	91.0	91.4	84.9	**95.5**	89.8	70.0	-	69.6	**90.0**	72.1
power drill	86.3	88.8	**92.6**	92.5	88.8	66.1	-	**86.1**	85.2	77.7
wood block*	83.9	75.7	84.3	**93.0**	90.7	83.9	-	84.3	**93.0**	90.7
scissors	80.8	75.2	**93.3**	80.9	83.0	51.0	-	**87.0**	71.2	59.7
large marker	**86.5**	81.2	84.9	85.2	74.9	74.6	-	76.6	**77.0**	63.9
large clamp*	80.3	88.6	92.0	**94.7**	78.3	80.3	-	92.0	**94.7**	78.3
extra large clamp*	74.1	83.5	**88.9**	80.7	54.7	74.1	-	**88.9**	80.7	54.7
foam brick*	92.1	81.3	90.7	**93.8**	89.9	92.1	-	90.7	**93.8**	89.9
MEAN	85.9	87.1	90.1	**91.2**	86.2	64.6	–	82.6	**83.3**	74.6

Table 5: Per-object results on YCB-V. * marks symmetric objects. We compare single-stage end-to-end DETR-based methods using AUC of ADD-S and ADD(-S), where available. Best results are **highlighted**; results in the range of 2% with the state of the art are highlighted in blue.

show that TinyDETR-Pose achieves competitive pose accuracy while being substantially more efficient than existing DETR-based single-stage pose-estimation approaches. In particular, our model obtains an AUC of ADD-S of **85.9** and achieves the best ADD(-S) accuracy at the $0.1d$ threshold with **93.6**, while requiring approximately **3.5**$\times$ fewer parameters than other DETR-based methods. Moreover, TinyDETR-Pose runs in real time and reaches an inference latency of only $\sim$4.5 ms per frame on an NVIDIA Jetson Nano using TensorRT. These results demonstrate that accurate transformer-based 6D pose estimation can be made practical for resource-constrained edge deployment.

Future Work & Limitations. Despite its efficiency and strong real-time performance, several directions remain open for future work. First, the evaluation should be extended beyond YCB-V to additional benchmarks such as LM-O [9] and to industrial datasets, where stronger occlusions, textureless objects, and domain-specific object geometries may further challenge the model. Second, pose-specific data augmentations, such as those used in YOLO-6D-Pose [24], could improve robustness to occlusion, truncation, illumination changes, and challenging object configurations. Another promising direction is the refinement and analysis of the Hungarian matcher. While our current matcher is designed to be symmetry-safe by relying on class and 2D spatial cues, more detailed studies could investigate occlusion-aware or pose-aware matching costs without introducing instability for symmetric objects. In addition, the number of object queries N remains an important hyperparameter. Since some YCB-V implementations operate with fewer queries, a dedicated query-count optimization could further reduce the model size and computational cost. Future work should also investigate symmetry-aware rotation representations, such as SARR [14], and analyze whether they can improve pose accuracy, especially for ambiguous or partially symmetric objects. Furthermore, although TinyDETR-Pose is already efficient on edge hardware, the AUC of ADD(-S) metric still leaves room for improvement. Additional edge-oriented optimizations such as quantization, pruning, distillation, or improved TensorRT deployment could further increase runtime efficiency while preserving accuracy. Finally, a more systematic analysis of pretrained weights and scaling behavior is needed. Current DETR-based architectures for 2D detection benefit strongly from large-scale pretraining, and similar large-scale pretraining on pose estimation datasets may substantially improve the representation quality of TinyDETR-Pose. Scaling the model in a controlled way, together with improved pretraining strategies, could further close the remaining accuracy gap while maintaining the efficiency advantages of the proposed lightweight design.

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